



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

STATISTICS
PAPER 2

6046/2
3 hours

SPECIMEN PAPER

Additional materials:
Answer booklet
Graph paper
List of Formulae MF 7
Non programmable Scientific calculator
Mathematical instruments

INSTRUCTIONS TO CANDIDATES

Write your Name, Centre number and Candidate number in the spaces provided on the answer booklet.

Answer **all** questions in **Section A** and any **five** from **Section B**.

If a numerical answer cannot be given exactly, and the accuracy required is not specified in the question, then in the case of an angle it should be given to the nearest degree, and in other cases it should be given correct to 2 significant figures.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

This specimen paper consists of 7 printed pages.

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Section A [40]
Answer all questions in this section

1. The following set of numbers depict the marks scored by students in a Sesotho test;
12; 15; 16; 19; 20; 24; 27; 28; 30.

Calculate the

- (a) mean,
- (b) variance,
- (c) standard deviation.

2. A volleyball committee of six members is to be chosen from a group of fifteen form 5 learners.

- (a) Calculate the number of ways the committee can be chosen.
- (b) Among the fifteen form 5 learners there is a pair of twin brothers and a pair of twin sisters.

Calculate the probability that both twin sisters and twin brothers will be chosen.

3. (a) State any **one** condition for a Poisson Distribution model.
- (b) The number of calls made to a Mutema Health Centre can be modelled by Poisson distribution with standard deviation of 2 calls per ten-minute interval.

Find the probability that at least 2 calls were made in 7.5 minutes.

4. M and N are two events such that $P(M) = 0.4$, $P(N) = 0.3$ and $P(M/N) = 0.5$

Calculate

- (a) $P(M \cap N)$,
- (b) $P(N/M)$.

5. The random variable X is normally distributed with mean μ and variance, δ^2 . It is given that $P(X < 95) = 0.68$ and $P(X > 130) = 0.19$. Calculate the value of μ and δ .

[5]

6. A bag contains ten identical balls of which five are black and seven are green. An experiment consists of drawing at random, three balls from the bag one at a time.

(a) Draw a tree diagram to show all the possible outcomes of the experiment.

[3]

(b) Hence or otherwise, find the probability that all three balls are of the same colour.

[2]

Ten students sat for a Mathematics test marked out of 50 and their marks were recorded as follows:

37	30	31	32	19
40	47	50	38	28

(a) Draw a stem and leaf diagram to illustrate the data.

[3]

(b) Find the mean mark.

[2]

The discrete random variable X has the probability distribution shown below:

x	1	2	3	4	5
$P(X = x)$	$\frac{1}{5}$	$\frac{1}{10}$	a	$\frac{2}{5}$	$\frac{1}{5}$

Find the

(a) value of a .

[2]

(b) expectation of X .

[3]

4
Section B [80 marks]
Answer any 5 questions from this section
Each question carries 16 marks

Mr Dhibha and Mrs Mudhibhisi were judges at a choir competition in which there were ten school choirs competing. The judges' marks in percentages are shown below.

Choir	Mr Dhibha (X)	Mrs Mudhibhisi (Y)
A	67	70
B	42	59
C	85	71
D	51	38
E	39	55
F	97	62
G	81	80
H	70	76
I	61	81
J	79	96

a) Plot a scatter diagram to illustrate the data. [4]

b) Taking Mr Dhibha's marks as x and Mrs Mudhibhisi's marks as y , calculate the equation of regression line of y on x . [5]

An eleventh school choir came very late for the competition. Mr Dhibha awarded the choir a mark of 65%. Mrs Mudhibhisi had already left when the eleventh choir arrived.

Estimate the mark that the eleventh school choir would have obtained from Mrs Mudhibhisi. [2]

(i) Find the product moment correlation coefficient (r). [3]

(ii) Compute the coefficient of determination (r^2). [2]

10. The teacher believes that the average mark for Pure Mathematics learners in the final examination is 75. A sample of 20 learners' examination mark is as follows:

80 68 72 73 76 81 76 73 65 50
63 71 70 70 72 73 65 68 72 73

- (a) Calculate the unbiased estimate of the

(i) mean,

[2]

(ii) variance,

[3]

(iii) standard deviation.

[2]

- (b) Assuming that the examination marks follow a Normal Distribution, test at 5% level of significance whether the teacher is overstating the average mark.

[9]

- (a) The table below shows a summary of Type I and Type II errors. Copy and complete the table below to show the outcome and errors associated with the Type I and Type II errors.

Test Decision		Accept H_0	Reject H_0
Actual Situation	H_0 is true	—	Type I error
	H_1 is false	—	correct

[2]

- (b) A random observation is taken from a Binomial Distribution $X \sim \text{Bin}(20; p)$ and used to test the null hypothesis $p = 0.6$. The critical region is chosen to be $X \geq 18$.

Find the probability of making a

(i) Type I error,

[3]

(ii) Type II error.

[4]

- (c) Mango seeds have a germination rate of 80%. Ten seeds were treated and 9 germinated.

Test at 5% significance level, whether this is evidence of an increase in the germination rate.

[7]

12. The probability of a bean seed germinating is 0.65.

Find the probability that in a random sample of 500 seeds.

- (a) less than 350 germinate, [6]
 (b) more than 310 germinate, [3]
 (c) between 320 and 340 germinate, [4]
 (d) not more than 250 seeds germinate. [3]

13. (a) List any **four** components of time series. [4]

- (b) The table below shows the number of winter jerseys in thousands produced by a factory in four seasons of the year for the period 2018 to 2021.

Season	2018	2019	2020	2021
Summer	60	70	53	48
Autumn	70	50	63	73
Winter	80	75	70	78
Spring	51	49	51	45

- (i) Draw a time series graph to illustrate the data. [5]
 (ii) Calculate the 5-point moving average. [4]
 (iii) Plot the 5-point moving average graph on the same axes as the graph in (i). [3]

The daily times, x , minutes for 100 pupils who walk daily to a secondary school in Gwelo Village are tabulated, below.

Time (minutes)	10-15	16-25	26-35	36-45	46-55
Number of pupils	20	10	15	25	30
Frequency density	3.1	a	1.5	b	3

- (a) Find the values of a and b . [4]
- (b) Draw a histogram to illustrate the daily times for the 100 pupils. [4]
- (c) Calculate the estimate of the
- (i) mean, [3]
- (ii) standard deviation. [5]
- (a) State any **one** situation in which a Chi-square significance test can be used. [1]
- (b) A Mathematics teacher wishes to determine whether there is a relationship between a group of students and the Mathematics Syllabus they prefer. The following data was obtained.

Group	New Syllabus	Old Syllabus	No preference	Total
Rural school learner	15	30	10	55
Urban school learner	21	40	14	75
Total	36	70	24	130

Test at 10% level of significance whether there is a difference in learners' Opinion, according to group. [15]